CHAPTER D7

Developer productivity and operational enablement

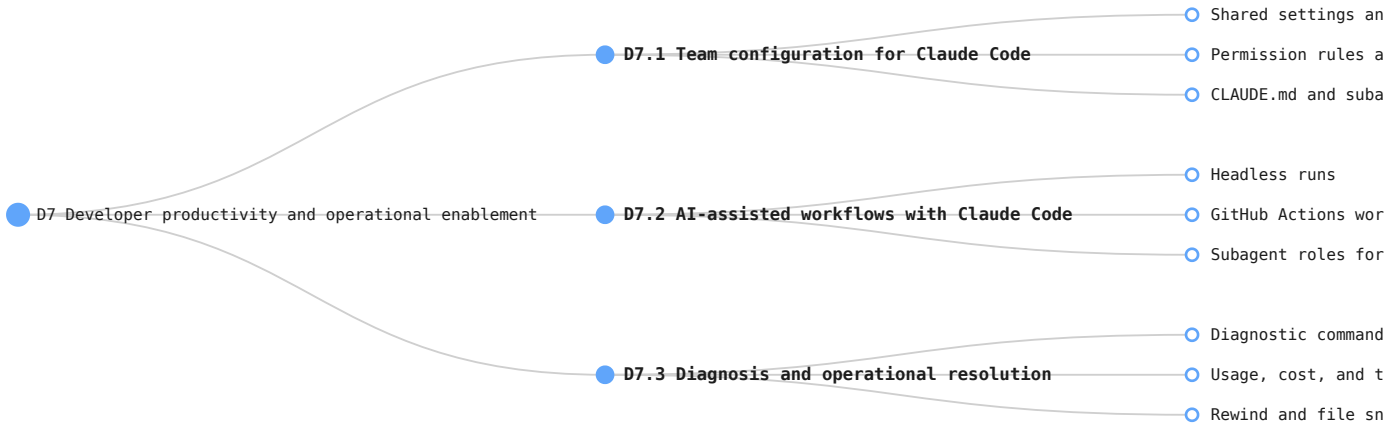
WEIGHT 7%. INFERENCE FROM SECONDARY SOURCES.

Settings and hooks enforce. CLAUDE.md suggests.

3 sections, 9 components, 9 recall cues.

CONCEPT MAP

Units branch to components. On a small screen, scroll sideways.



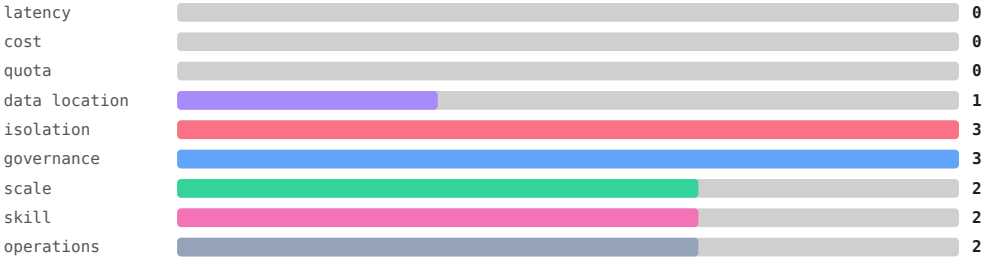
SECTIONS

D7.1 CHAPTER D7, DEVELOPER PRODUCTIVITY AND OPERATIONAL ENABLEMENT

Team configuration for Claude Code

Control: permission rule or hook. Convention: CLAUDE.md.

CONSTRAINT FAMILIES



SHARED SETTINGS AND SCOPES

One committed settings file gives every clone the same permissions and hooks, and personal files hold the exceptions.

Rule Use a permission rule or a hook for a control, and use CLAUDE.md for a convention.

Trap A CLAUDE.md note that asks the agent not to read secret files. The docs call CLAUDE.md context, not enforced configuration, so only a deny rule blocks the read.

Exam tip If the scenario needs a block, choose a deny rule, a PreToolUse hook, or managed settings.

PERMISSION RULES AND HOOKS

Permission rules and hooks decide a tool call in code, so the result does not depend on what the model chooses.

Rule Use a deny rule to block a tool and a hook to run a check at a lifecycle event.

Trap A PreToolUse hook that returns allow, which looks like approval. A deny rule still blocks the call, so the hook does not override the team policy.

Exam tip If the stem says the model must not run a command, choose the deny rule over a note in the memory file.

CLAUDE.MD AND SUBAGENT GUIDANCE

Guidance files shape how the model works, and they cannot stop a tool call.

Rule Write a convention in CLAUDE.md, and write a required block in a permission rule.

Trap A bigger model with a long CLAUDE.md file, expected to enforce a secret rule. Adherence to guidance is not a lock, so the rule belongs in permissions.

Exam tip If an answer relies on the model to follow a note for safety, reject it and look for a deny rule.

▼ RECALL CUES (3)

1. **stop the agent from opening secret files on every clone**
   Answer deny rules in the shared settings file
   Rule Put the deny rule in the team file. A note in CLAUDE.md does not block a read.
   - A CLAUDE.md note that asks the agent not to read secrets
   - A deny rule in one developer local file only
2. **one developer needs an exception to the team permissions**
   Answer local settings override
   Rule Put the personal exception in the local file. A deny rule from any scope still applies.
   - A change to the committed shared file for one person
   - A user-level allow rule that expects to override a project deny
3. **every edit must run the lint command**
   Answer PostToolUse hook in the shared settings file
   Rule Use a hook for a step that must run after each edit.
   - A CLAUDE.md line that asks the model to run lint
   - A PreToolUse hook used for a step that runs after the edit

▼ NUMBERS AND VERSIONS (6)

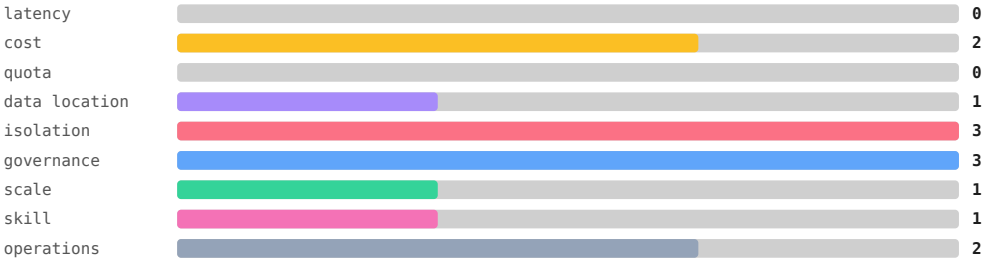
- A PreToolUse hook that exits with code 2 blocks the tool call before permission rules run. source
- The default timeout for a command hook is 600 seconds. source
- Keep each CLAUDE.md file under 200 lines. source
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D7.2 CHAPTER D7, DEVELOPER PRODUCTIVITY AND OPERATIONAL ENABLEMENT

AI-assisted workflows with Claude Code

--tools restricts. --allowedTools only auto-approves.

CONSTRAINT FAMILIES



HEADLESS RUNS

A print mode run returns a result and an exit status, so a script can decide whether the job passes.

Rule Restrict each scripted run with --tools, approve only those tools, and check the exit status before the job passes.

Trap An --allowedTools list read as a tool restriction. The flag only auto-approves the listed tools. To restrict which tools exist, use --tools, and bypass mode belongs only in isolated environments.

Exam tip If the stem asks to restrict tools in a CI run, look for --tools or a deny rule, not --allowedTools alone.

GITHUB ACTIONS WORKFLOWS

A workflow trigger starts Claude Code in the repository, and repository secrets hold the credential.

Rule Use the GitHub Action for a repository trigger, and use an API key for a secret that several repositories share.

Trap A personal subscription token stored as a shared organization secret. The docs say the token is tied to the subscription of the person who created it, so an API key fits a shared secret.

Exam tip If the stem names a shared secret across repositories, choose an API key over an OAuth token.

SUBAGENT ROLES FOR REVIEW AND EXPLORATION

A subagent with a narrow tools list can explore or review without the power to change files.

Rule Set the tools field on each role to the smallest set that the role needs.

Trap A review subagent with no tools field, which inherits write tools. The role then can edit files that a review should only read.

Exam tip If a stem asks for a read-only reviewer, look for a tools list that omits write tools.

▼ RECALL CUES (3)

1. **run a model check in CI and fail the job on error**
   Answer print run with an exit status check
   Rule Use -p, restrict tools with --tools, approve only those tools, and read the exit status.
   - An interactive session in CI
   - Bypass mode with no tools list
   - An --allowedTools list as the only restriction
2. **a script needs fields from the model output**
   Answer JSON output with a schema
   Rule Use --output-format json with --json-schema when the script parses fields.
   - Parsing free text with a pattern match
   - A request for JSON in the prompt only
3. **a comment on an issue starts a code change**
   Answer GitHub Action with repository secret
   Rule Use the GitHub Action for a comment trigger and keep the credential in a repository secret.
   - A developer script run by hand
   - A personal subscription token as a shared secret

▼ NUMBERS AND VERSIONS (2)

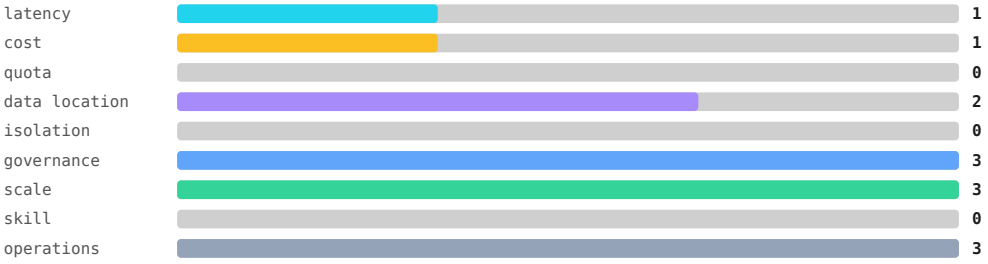
- Claude Code exits with code 0 on success and a non-zero code when the run fails. source
- Piped stdin is capped at 10MB. source

D7.3 CHAPTER D7, DEVELOPER PRODUCTIVITY AND OPERATIONAL ENABLEMENT

Diagnosis and operational resolution

Run the built-in check for the symptom first.

CONSTRAINT FAMILIES



DIAGNOSTIC COMMANDS

Read the loaded state before you change a setting, a hook, or a server, because a file on disk can differ from what loaded.

Rule Run the built-in check that matches the symptom before you change the setting or the code.

Trap A CLAUDE.md edit meant to make a hook run. Hooks run from settings files, and a guidance note never starts a hook, so the check is the hook configuration and the debug log.

Exam tip If a setting seems ignored, check the settings sources with /status before you edit the file.

USAGE, COST, AND TELEMETRY

Cost and latency changes need per-session numbers before anyone changes the model.

Rule Read the cost and token data for the affected sessions before you change the model.

Trap A smaller model to cut cost before the team reads the usage data. The change may miss the cause and can lower answer quality.

Exam tip If a cost question asks for the first step, choose the usage read over the model change.

REWIND AND FILE SNAPSHOTS

Rewind restores file edits that Claude Code tracked, so a bad change does not need a manual undo.

Rule Use rewind for tracked Claude edits and version control for everything else.

Trap A belief that rewind undoes every change in the session. Rewind does not track Bash changes to files that Claude did not edit.

Exam tip If a stem says a shell command changed a file, look for version control in the answer, not rewind.

▼ RECALL CUES (3)

1. **a hook that should run does not run**
   Answer loaded config and debug log check
   Rule Check the hook in /hooks, then start the session with claude --debug.
   - A CLAUDE.md note that asks for the hook
   - A larger model to fix a hook
2. **cost rises after a prompt change**
   Answer per-session usage read first
   Rule Read the usage and token data, then find the context that grew.
   - A smaller model with no data
   - A longer cache lifetime with no check
3. **an agent edit must be undone after a shell command changed a file**
   Answer version control revert for the Bash change
   Rule Use rewind for tracked edits and version control for Bash changes.
   - Rewind alone for a Bash-only change
   - A manual edit with no record

▼ NUMBERS AND VERSIONS (5)

- Across enterprise deployments, the average cost is around $13 per developer per active day. source
- The default export interval for metrics is 60000 milliseconds. source
- Claude Code keeps file snapshots for the 100 most recent checkpoints in a session. source
- Claude Code deletes a session's file snapshots about 30 days after the session last saved one. source
- The cleanupPeriodDays setting sets the retention period, with a default of 30 days. source